NEKHOROSHEV STABILITY FOR QUASI-CONVEX HAMILTONIANS: BLUEPRINT (TRACK N)

SCOTT ARMSTRONG AND ZOHAR KOMARGODSKI

Abstract. A proof plan, step by step, for the three frozen statements of track N of the KAM formalization: the exponential stability estimate for strictly quasi-convex nearly integrable Hamiltonians with a complex Hessian bound (N-HEAD-C, after Pöschel 1993, Theorem 1), the exact integrable dynamics for zero perturbation (N-ZERO) and the nonsingular isoenergetic derivative (N-QC). Every result is labelled and cites its pinned source. This version contains Part 0 only; the plan of the remaining parts is in `PLAN.md`.

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1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the three frozen theorems of track N (owner approval D20, 27 October 2026) and the eleven frozen definitions of `KAM/Frozen/Nekhoroshev/Defs/` together with the five track-K definitions they use. The Lean declarations are authoritative; the statements below are transcriptions of them, with the Lean names recorded under each. N-HEAD with real bounds only is *not* frozen (source gap F-N1, D17/D18/D20); the frozen headline is N-HEAD-C. The additions of D22 (N-PARTIAL-CONVEX, N-AVERAGING) are not yet frozen and are not part of this part. Results of track K's blueprint are cited as [K:`bp:p1:...`].

1.1. Conventions.

- $n \in \mathbb{N}$ is the number of degrees of freedom; N-HEAD-C assumes $n \geq 3$ (D17), N-ZERO and N-QC are stated for every n. A phase point is $z = (\theta, I)$ with angles θ and actions I, in this order.

- Ambient spaces are those of the Lean text: $\mathbb{R}^n = (\texttt{Fin}\ n \to \mathbb{R})$, $\mathbb{C}^n = (\texttt{Fin}\ n \to \mathbb{C})$, with the max norms $|v| = \max_i |v_i|$ on both. The embedding $\mathbb{R}^n \subset \mathbb{C}^n$ is componentwise. The norm $\|\cdot\|$ in every frozen statement is the max norm, with two exceptions that are Euclidean on purpose: orthogonality in quasi-convexity uses the dot product $\sum_i \omega_i v_i$, and the phase space of a solution is $\mathbb{R}^n \times \mathbb{R}^n$.
- Operator norms of linear and multilinear maps are the induced norms of these max norms. In particular $\|D^2 h(z)\|$ for complex h is the norm of $D^2 h(z) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ (Lean `fderiv` applied twice), i.e. $\sup_{|u|,|v|\le 1} |D^2 h(z)[u, v]|$.
- Angles are *not* reduced modulo $\mathbb{Z}^n$ in the statements: $\theta \in \mathbb{R}^n$ or $\theta \in T_s \subset \mathbb{C}^n$, and 1-periodicity is a hypothesis on f (Definition 1.12). So the torus is $\mathbb{R}^n/\mathbb{Z}^n$ and the Fourier characters are $e^{2\pi i k\cdot\theta}$ as in track K ([K:`bp:p1:decay`], [K:`bp:p1:summation`]). Pöschel's torus is $\mathbb{R}^n/2\pi\mathbb{Z}^n$ and his norms are Euclidean; the conversion is a separate step of the proof (Part 1 of `PLAN.md`).
- The complex strip is $T_s = \{\theta \in \mathbb{C}^n : |\operatorname{Im}\theta_i| < s\ \forall i\}$ (Lean `KAM.complexStrip`), as in track K.

1.2. **Track-K definitions used.**

Definition 1.1 (Complex action neighbourhood)**.** For $B \subset \mathbb{R}^n$ and $r \in \mathbb{R}$, $B_r = \{I \in \mathbb{C}^n : \exists J \in B,\ |I - J| < r\}$ (max norm, J viewed in $\mathbb{C}^n$). Thus $B_r = \emptyset$ for $r \le 0$, and for B open B_r is open.
Frozen Lean declaration: `KAM.actionNbhd`
Source: BM11 p. 4 (D_s); track K, [K:`bp:p0:domains`].

Definition 1.2 (Frequency map)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $I \in \mathbb{R}^n$, $\omega_h(I) = \nabla h(I) \in \mathbb{R}^n$ is the vector with $\omega_h(I)_i = D\big(J \mapsto \operatorname{Re} h(J)\big)(I)[e_i]$, the real partial derivatives of $\operatorname{Re} h$ on $\mathbb{R}^n$ at I. It is 0 if $\operatorname{Re} h|_{\mathbb{R}^n}$ is not differentiable at I (Lean totality, Remark 1.18).
Frozen Lean declaration: `KAM.frequencyMap`
Source: BM11 p. 1; Pos93 p. 190 ($\omega(I) = \partial_I h(I)$).

Definition 1.3 (Hamiltonian vector field)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and $z = (\theta, I)$, $X_H(z) = \big((\partial_{I_i} H(z))_i,\ (-\partial_{\theta_i} H(z))_i\big)$, where $\partial_{I_i} H(z) = DH(z)[(0, e_i)]$ and $\partial_{\theta_i} H(z) = DH(z)[(e_i, 0)]$ are the Fréchet derivatives in the Lean sense (0 if H is not differentiable at z).
Frozen Lean declaration: `KAM.hamiltonianVectorField`
Source: BM11 p. 1; track K, [K:`bp:p0:hamiltonian`].

Definition 1.4 (Real Hamiltonian)**.** For $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and real $z = (\theta, I)$, $H_{h,f}(z) = \operatorname{Re}\big(h(I) + f(\theta, I)\big)$, with θ, I viewed in $\mathbb{C}^n$.
Frozen Lean declaration: `KAM.realHamiltonian`
Source: BM11 p. 5 ($H = h + f$).

Remark 1.5 (Strip and neighbourhood notation). `KAM.complexStrip` $n\, s = T_s$ as above; it is empty for $s \le 0$.
Frozen Lean declaration: `KAM.complexStrip`

1.3. **Frozen track-N definitions.**

Definition 1.6 (Action ball)**.** $B(0, R) = \{I \in \mathbb{R}^n : |I| < R\}$, the open ball of the max norm; written `actionBall` $n\, R$.
Frozen Lean declaration: `KAM.Nekhoroshev.actionBall`
Source: BM11 p. 4.

Definition 1.7 (Complex domain)**.** $D_s(R) = T_s \times B(0, R)_s \subset \mathbb{C}^n \times \mathbb{C}^n$; `complexDomain` $n\, R\, s$. It is the period-one lift of BM11's D_s, with a single width s for angles and actions. It is open.
Frozen Lean declaration: `KAM.Nekhoroshev.complexDomain`
Source: BM11 p. 4 (D_s); ours for the lift.

Definition 1.8 (Real action domain). $\mathcal{B}_s(R) = \{I \in \mathbb{R}^n : I \in B(0,R)_s\}$; `realActionDomain` $n\,R\,s$.
Frozen Lean declaration: `KAM.Nekhoroshev.realActionDomain`
Source: BM11 p. 4 (B_s).

Remark 1.9 (Computing $\mathcal{B}_s(R)$). Since $|I-J|$ for real I, J is the real max norm, $\mathcal{B}_s(R) = \{I \in \mathbb{R}^n : \exists J,\ |J| < R,\ |I-J| < s\} = B(0, R+s)$ for $R, s > 0$. This identity is ours and is proved from the definitions (it is not stated in the source); it shows that $B(0,R) \subset \mathcal{B}_s(R)$ and that $\mathcal{B}_s(R)$ is open.

Definition 1.10 (Real part of the integrable part). For $h : \mathbb{C}^n \to \mathbb{C}$ and $I \in \mathbb{R}^n$, $h_{\mathbb{R}}(I) = \operatorname{Re} h(I)$; `realPart` h.
Frozen Lean declaration: `KAM.Nekhoroshev.realPart`
Source: BM11 p. 4 ($h : B_s \to \mathbb{R}$).

Definition 1.11 (Analytic integrable part, $h \in A_s(D)$). `IsAnalyticIntegrable` $R\,s\,h$ holds if

(i) h is complex differentiable on $B(0,R)_s$;
(ii) h is continuous on $\overline{B(0,R)_s}$ (closure in $\mathbb{C}^n$);
(iii) there is $A \geq 0$ with $|h(I)| \leq A$ for all $I \in B(0,R)_s$;
(iv) $\operatorname{Im} h(I) = 0$ for all $I \in \mathcal{B}_s(R)$;
(v) $h_{\mathbb{R}}$ is of class C^3 on $\mathcal{B}_s(R)$ (real Fréchet sense; the set is open).

No numerical bound on h is fixed. Clause (v) is redundant by holomorphy ([K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:real-restriction`]) and records the regularity that guards every real derivative.
Frozen Lean declaration: `KAM.Nekhoroshev.IsAnalyticIntegrable`
Source: BM11 p. 4 ($A_s(D)$, integrable case, with continuity on the closure).

Definition 1.12 (Analytic perturbation of size ε). `IsAnalyticPerturbation` $R\,s\,\varepsilon\,f$ holds if $\varepsilon \geq 0$ and

(i) f is complex differentiable on $D_s(R)$ and continuous on $\overline{D_s(R)}$;
(ii) $f(\theta + k, I) = f(\theta, I)$ for all $(\theta, I) \in D_s(R)$ and $k \in \mathbb{Z}^n$;
(iii) $\operatorname{Im} f(\theta, I) = 0$ for all $\theta \in \mathbb{R}^n$ and $I \in \mathcal{B}_s(R)$;
(iv) $|f(z)| \leq \varepsilon$ for all $z \in D_s(R)$ (complex absolute value).

The size is measured by $\leq \varepsilon$, not the source's strict bound; the source is applied with 2ε. For $\varepsilon = 0$ this forces $f = 0$ on $D_s(R)$.
Frozen Lean declaration: `KAM.Nekhoroshev.IsAnalyticPerturbation`
Source: BM11 pp. 4–5 ($f \in A_s(D)$, $|f|_s < \varepsilon$); deviation: $\leq \varepsilon$ (ruling Q2, D18).

Definition 1.13 (Derivative bounds $B(M)$). For $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$ and $M \in \mathbb{R}$, `HasDerivativeBounds` $U\,h\,M$ holds if $M > 0$, U is open, $h_{\mathbb{R}} \in C^3(U)$, and for all $I \in U$, all $k \in \{1,2,3\}$ and all $(i_1, \ldots, i_k) \in \{1, \ldots, n\}^k$,

$$\left|D^k h_{\mathbb{R}}(I)[e_{i_1}, \ldots, e_{i_k}]\right| \leq M.$$

These are all ordered coordinate partials of orders one to three; for C^3 functions they coincide with the multi-index partials $\partial^k h$ of the source (symmetry, [K:`bp:p1:ext-schwarz`]). The bound is on coordinate partials, not on operator norms.
Frozen Lean declaration: `KAM.Nekhoroshev.HasDerivativeBounds`
Source: BM11 p. 4, $B(M)$.

Definition 1.14 (Strict quasi-convexity $QC(m)$). For $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$, $m \in \mathbb{R}$, `IsStrictlyQuasiConvex` $U\,h\,m$ holds if $m > 0$, U is open, $h_{\mathbb{R}} \in C^2(U)$, and for every $I \in U$: $\omega_h(I) \neq 0$ and

$$\sum_i \omega_h(I)_i v_i = 0 \implies m|v|^2 \leq D^2 h_{\mathbb{R}}(I)[v,v] \qquad (v \in \mathbb{R}^n),$$

with $|v|$ the max norm on the left and the Euclidean dot product in the hypothesis.

Frozen Lean declaration: `KAM.Nekhoroshev.IsStrictlyQuasiConvex`
Source: BM11 p. 4, $QC(m)$ (the source's norm on the right is the max norm, orthogonality is Euclidean).

Definition 1.15 (Complex Hessian bound)**.** `HasComplexHessianBound` $R\,s\,h\,M_{\mathbb{C}}$ holds if $M_{\mathbb{C}} > 0$, h is C^2 in the complex sense on $B(0,R)_s$ and $\|D^2h(z)\| \le M_{\mathbb{C}}$ for all $z \in B(0,R)_s$, the norm being that of $D(Dh)(z) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ for the max norm. Every entry $|\partial_i\partial_j h|$ is then $\le M_{\mathbb{C}}$, and the Euclidean operator norm of the Hessian matrix is $\le nM_{\mathbb{C}}$. This is not the spectral radius of a complex symmetric matrix.
Frozen Lean declaration: `KAM.Nekhoroshev.HasComplexHessianBound`
Source: Pos93 p. 190, "The Hessian of the integrable hamiltonian h" (conversion by n recorded in `docs/tracks/N/sources.md`, anchor POS190-HESS).

Definition 1.16 (Solutions on a time set)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $U \subset \mathbb{R}^n$, $J \subset \mathbb{R}$ and $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$, `IsSolutionOn` $H\,U\,J\,z$ holds if for every $t \in J$: $z(t)_2 \in U$, H is differentiable at $z(t)$, and z is differentiable at t (two-sided, also at endpoints of J, since z is defined on $\mathbb{R}$) with $z'(t) = X_H(z(t))$.
Frozen Lean declaration: `KAM.Nekhoroshev.IsSolutionOn`
Source: BM11 Theorem 2.1 p. 5 and the maximal-solution argument p. 13.

Definition 1.17 (The integrable orbit)**.** For $h : \mathbb{C}^n \to \mathbb{C}$, $\theta_0, I_0 \in \mathbb{R}^n$ and $t \in \mathbb{R}$: $\gamma_{\theta_0,I_0}(t) = (\theta_0 + t\,\omega_h(I_0),\, I_0)$.
Frozen Lean declaration: `KAM.Nekhoroshev.integrableOrbit`
Source: BM11 p. 1 ($H(\theta, I) = h(I)$).

Remark 1.18 (Lean totality). The frozen definitions and theorems use `fderiv` (which is 0 at a point of non-differentiability), `iteratedFDeriv`, the operator norm, `rpow` ($\varepsilon^{\pm 1/(2n)}$), `Real.exp` and the max norm; there is no matrix inverse, no Bochner integral and no division by a quantity that may vanish. The guards are:

(a) ω_h and $D^2h_{\mathbb{R}}$ are used only on the open set $\mathcal{B}_s(R)$ where $h_{\mathbb{R}}$ is C^3 (Definitions 1.11(v), 1.13, 1.14), so the derivatives are the genuine ones.
(b) D^2h for complex h is used only on $B(0,R)_s$, open, where h is C^2 (Definition 1.15).
(c) X_H is used only inside `IsSolutionOn`, which requires differentiability of H at every visited point; in N-ZERO and N-HEAD-C this is a proof obligation: $H_{h,f}$ is real-differentiable on $\mathcal{B}_s(R) \times \mathbb{R}^n \supset B(0,R) \times \mathbb{R}^n$ because h, f are holomorphic on open sets containing the real points ([K:`bp:p1:hamiltonian-real`]).
(d) $\varepsilon^{-1/(2n)}$ and $\varepsilon^{1/(2n)}$ occur only with $\varepsilon > 0$ in N-HEAD-C, and $2n \neq 0$.
(e) In N-QC the map $\Psi(I', \lambda) = (h_{\mathbb{R}}(I'), \lambda\omega_h(I'))$ is differentiable at (I, λ) because $h_{\mathbb{R}} \in C^2$ on the open set $U \ni I$; `fderiv` is then the true derivative.

1.4. **The frozen theorems.**

Theorem 1.19 (N-HEAD-C: exponential stability, complex Hessian bound)**.** *Let $n \ge 3$ and $R, s, M, m, M_{\mathbb{C}}, \omega_{\min} > 0$. Then there are $C, T, c, \varepsilon_0 > 0$ with*

$$C\,\varepsilon_0^{1/(2n)} \le R/4$$

such that the following holds for every $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and $\varepsilon \in \mathbb{R}$ with

(H1) `IsAnalyticIntegrable` $R\,s\,h$ *(Definition 1.11);*
(H2) `HasComplexHessianBound` $R\,s\,h\,M_{\mathbb{C}}$ *(Definition 1.15);*
(H3) $\omega_{\min} \le |\omega_h(I)|$ *for every $I \in B(0,R)$ (max norm);*
(H4) `IsStrictlyQuasiConvex` $\mathcal{B}_s(R)\,h\,m$ *(Definition 1.14);*
(H5) `HasDerivativeBounds` $\mathcal{B}_s(R)\,h\,M$ *(Definition 1.13);*
(H6) `IsAnalyticPerturbation` $R\,s\,\varepsilon\,f$ *(Definition 1.12);*
(H7) $0 < \varepsilon \le \varepsilon_0$.

For every $\theta_0 \in \mathbb{R}^n$ and $I_0 \in B(0, R/2)$, with $\tau := T\exp\bigl(c\,\varepsilon^{-1/(2n)}\bigr)$ and $H = H_{h,f}$, there is $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ with

(a) $z(0) = (\theta_0, I_0)$;
(b) `IsSolutionOn` $H\, B(0,R)\, [-\tau, \tau]\, z$;
(c) $|z(t)_2 - I_0| \le C\varepsilon^{1/(2n)}$ for all t with $|t| \le \tau$;
(d) (uniqueness) for every $L \in [0, \tau]$ and every $y : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ with $y(0) = (\theta_0, I_0)$ and `IsSolutionOn` $H\, B(0,R)\, [-L, L]\, y$: $y(t) = z(t)$ for all $t \in [-L, L]$.

The order of choice is: n; $R, s, M, m, M_{\mathbb{C}}, \omega_{\min}$; then C, T, c, ε_0, depending only on these seven data; then h, f, ε, then θ_0, I_0; the solution z may depend on all of them. There is no dependence on A in Definition 1.11(iii). In (H4)–(H5) the set is $\mathcal{B}_s(R) = B(0, R+s)$ (Remark 1.9), in (H3) the smaller ball $B(0, R)$.

Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_nekhoroshev_complexBound`

Source: Pos93 Theorem 1, pp. 190–191 (with the conversions and anchors of `docs/tracks/N/sources.md`); exponent $1/(2n)$ as BM11 p. 5 with $\delta = (2n(n-1))^{-1}$; ruling D18 Q2, Q4; deviation D20: the extra inputs $M_{\mathbb{C}}$ and $\omega_{\min}$.

Source gap: The printed theorem is for a 2π-periodic torus, Euclidean norms and exceptional initial frequencies $\|\omega(I_0)\| \le mr_0/8$; the conversion to the frozen statement (rescaling $H \mapsto 2\pi H(\varphi/2\pi, I)$, choice of r_0, s_0 from the seven data, exclusion of the exceptional branch by $\omega_{\min}$, existence and uniqueness of the solution) is given only in `sources.md` and is proved in Part 10 of `PLAN.md`.

Theorem 1.20 (N-ZERO: exact integrable dynamics)**.** *Let $n \in \mathbb{N}$, $R, s > 0$, h with `IsAnalyticIntegrable` $R\, s\, h$ and f with `IsAnalyticPerturbation` $R\, s\, 0\, f$. For every $\theta_0 \in \mathbb{R}^n$ and $I_0 \in B(0, R/2)$, with $\gamma = \gamma_{\theta_0, I_0}$ (Definition 1.17):*

$$\gamma(0) = (\theta_0, I_0), \qquad \texttt{IsSolutionOn}\, H_{h,f}\, B(0,R)\, \mathbb{R}\, \gamma, \qquad \gamma(t)_2 = I_0 \quad (t \in \mathbb{R}).$$

Frozen Lean declaration: `KAM.Nekhoroshev.zero_perturbation`

Source: Ours (supplement to BM11 Theorem 2.1, p. 5, whose strict bound excludes $\varepsilon = 0$); from $H = h(I)$, BM11 p. 1.

Theorem 1.21 (N-QC: nonsingular isoenergetic derivative)**.** *Let $n \in \mathbb{N}$, $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$, $m \in \mathbb{R}$ with `IsStrictlyQuasiConvex` $U\, h\, m$. For $I \in U$ and $\lambda > 0$, the derivative at (I, λ) of*

$$\Psi : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n, \qquad \Psi(I', \lambda') = \bigl(h_{\mathbb{R}}(I'),\ \lambda'\omega_h(I')\bigr)$$

is a bijective linear map $\mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n$. Explicitly it is $D\Psi(I,\lambda)[v,u] = \bigl(\omega_h(I)\cdot v,\ u\,\omega_h(I) + \lambda D^2 h_{\mathbb{R}}(I)v\bigr)$. No uniform radius of invertibility is asserted.

Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_isoenergetic_derivative`

Source: BM11 Lemma 3.3 and its proof, pp. 9–10 (only the derivative assertion).

2. Hamiltonian ODE infrastructure (shared by tracks I, N, P)

This part is the single ODE layer of the three blueprints I, N, P. It has two levels. The *general layer* (§2.2–§2.6) is about an autonomous vector field F of class C^1 on an open subset W of a finite-dimensional real normed space E: local existence, uniqueness on *every* interval, maximal solutions, continuation (compact sets, and closed sets with uniform bounds), first exit from an open set. The *Hamiltonian layer* (§2.7) adds $F = X_H$: smoothness of X_H, energy conservation, translation in the angles. §2.8 shows that the frozen solution predicates of the three tracks (`IsSolutionOn`, `IsHamiltonianSolutionOn`, `IsRestrictedSolutionOn`, `IsNewtonSolution`, `IsFixedSolution`) are instances of the general predicate, and §2.9 gives the six track-N results of `PLAN.md`. Tracks I and P cite the general results as [N:`bpN:p3:gen-...`].

2.1. **Conventions and external facts.**

- E is a finite-dimensional real normed space (hence complete and proper: closed bounded sets are compact), $F : E \to E$ a *total* function, $W \subset E$ open. In the applications $E = \mathbb{R}^n \times \mathbb{R}^n$ with the max norm $|(\theta, I)| = \max(|\theta|, |I|)$ (the product norm of the Lean text, equal to the max norm on the $2n$ coordinates), or $E = (\mathbb{R}^2)^N \times (\mathbb{R}^2)^N$ for the Newtonian problems.
- F is C^1 *on* W means `ContDiffOn` $\mathbb{R}$ 1 F W; as W is open this gives $F \in C^1$ at each point of W, and no information off W. Off W the values of F never enter (Lean totality, below).
- An *interval* is an order-connected subset of $\mathbb{R}$ (Lean `OrdConnected`); open, closed, half-open, bounded or not, possibly a single point. $\overline{B}(x, \delta)$ is the closed ball.
- All derivatives $\dot{z}(t)$ in solution predicates are two-sided (Lean `HasDerivAt`) at every $t \in J$, including endpoints of J, exactly as in the frozen predicates ("z is defined on $\mathbb{R}$").

External facts. Mathlib-level statements, with the declarations that exist (checked in `Mathlib/Analysis/ODE/`); what is *not* in Mathlib is listed after them.

(E1) *Picard–Lindelöf, quantitative form.* Let $F : E \to E$, $x \in E$, $a, M, L \geq 0$ with $\|F(y)\| \leq M$ and F L-Lipschitz on $\overline{B}(x, a)$, $T \geq 0$ with $MT \leq a$. Then there is $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$ and α has derivative $F(\alpha(t))$ within $[t_0 - T, t_0 + T]$ at each t of that interval. (Mathlib: `ODE.IsPicardLindelof` with $r = 0$, time-independent field, and `IsPicardLindelof.exists_eq_forall_mem_Icc_hasDerivWithinAt_0`; the time constraint is the field `mul_max_le`: $M \max(t_{\max} - t_0, t_0 - t_{\min}) \leq a - r$. The contraction on iterates means no smallness of LT is needed.) Exists.

(E2) *Uniqueness with a Lipschitz constant on a set.* If $S \subset E$ and F is L-Lipschitz on S, $f, g : [a, b] \to E$ are continuous, have right derivative $F(f(t))$, $F(g(t))$ on $[a, b)$, take values in S and $f(a) = g(a)$, then $f = g$ on $[a, b]$; and the mirror statement with $f(b) = g(b)$ and left derivatives. (Mathlib: `ODE_solution_unique_of_mem_Icc_right`, `ODE_solution_unique_of_mem_Icc_left`.) Exists.

(E3) C^1 *implies locally Lipschitz.* If F is C^1 at p there are L and a neighbourhood on which F is L-Lipschitz (Mathlib: `ContDiffAt.exists_lipschitzOnWith`). Exists.

(E4) *Mean value inequality*: [K:`bp:p1:ext-mvt`]; for a differentiable map on a convex set with $\|DF\| \leq L$, F is L-Lipschitz there; for a function of one variable with derivative bounded by M on an interval, $\|\alpha(u) - \alpha(s)\| \leq M|u - s|$. Exists.

(E5) *Topology.* Heine–Borel in E (`ProperSpace`); continuous images of compact sets are compact; sequential compactness; $x \mapsto \operatorname{dist}(x, A)$ is continuous and positive on $E \setminus A$ for A closed (`Metric.infDist`). Exists.

(E6) *Calculus.* Chain rule `HasFDerivAt.comp_hasDerivAt`; a function continuous on $[s, t]$ with right derivative 0 on $[s, t)$ is constant (`constant_of_has_deriv_right_zero`); two one-sided derivatives agree ⇒ two-sided one (`HasDerivWithinAt.union` with $(-\infty, b] \cup [b, \infty) = \mathbb{R}$ and `hasDerivWithinAt_univ`); `ContDiffOn.fderiv_of_isOpen`: if H is C^{k+1} on an open set then DH is C^k there. Exists.

Not in Mathlib (checked: no maximal-solution, continuation, first-exit or uniform-existence-time statement in `Analysis/ODE`): everything of §2.2–§2.6 beyond (E1)–(E3), notably Lemmas 2.4, 2.8, Proposition 2.9, Theorems 2.11, 2.13, 2.14, and the Hamiltonian layer. They must be proved from the blueprint below.

Remark 2.1 (Lean totality). `KAM.hamiltonianVectorField` H uses `fderiv`, which is 0 at a point of non-differentiability, and the Newtonian fields use $1/\ell^3$ with $\ell =$ `planeLength` or `radius`. In every use below these are guarded: X_H occurs only on an open set where H is C^2 (or inside the frozen predicates, which assert differentiability at every visited point), and $\ell > 0$ at every visited point because the frozen predicates require it. The general layer never uses F off W.

2.2. **The general solution predicate; Lipschitz data.**

Definition 2.2 (Solutions in W)**.** For $F : E \to E$, $W \subset E$, $J \subset \mathbb{R}$ and $z : \mathbb{R} \to E$: `IsSolIn` $F\, W\, J\, z$ holds if for every $t \in J$, $z(t) \in W$ and z has derivative $F(z(t))$ at t (two-sided). Suggested Lean name `KAM.ODE.IsSolutionIn`; it is *not* a frozen declaration.
Source: Standard; shape dictated by the frozen predicates of Definition 1.16 and §2.8.

Elementary consequences (each a one-line Lean lemma, (E6)): (i) restriction: `IsSolIn` $F\, W\, J\, z$ and $J' \subset J$ give `IsSolIn` $F\, W\, J'\, z$; (ii) z is continuous on J (differentiable at each point of J); (iii) *local nature*: if J' is a neighbourhood of each of its points and $z = y$ on it, then `IsSolIn` $F\, W\, J'\, y \Leftrightarrow$ `IsSolIn` $F\, W\, J'\, z$ for open J'.

Lemma 2.3 (Symmetries)**.** *Let* `IsSolIn` $F\, W\, J\, z$.

(a) (time shift) For $c \in \mathbb{R}$, $t \mapsto z(t + c)$ is a solution on $J - c$.
(b) (time reversal) $t \mapsto z(-t)$ is a solution of $-F$ on $-J$, and $-F$ is C^1 on W if F is.
(c) (space translation) If $v \in E$ satisfies $W + v = W$ and $F(x + v) = F(x)$ for $x \in W$, then $t \mapsto z(t) + v$ is a solution of F in W on J.

Source: Standard.

Proof. **Step S2.1: Chain rule.** (a) and (b) are the chain rule for $t \mapsto t + c$, $t \mapsto -t$ ((E6)); (c): $z + v$ has the same derivative $F(z(t)) = F(z(t) + v)$, and $z(t) + v \in W + v = W$. □

Lemma 2.4 (Bounds and Lipschitz constants on compact sets)**.** *Let F be C^1 on the open set W and $K \subset W$ compact. Then F is bounded on K and there is $L \geq 0$ with $\|F(x) - F(y)\| \leq L\|x - y\|$ for $x, y \in K$.*
Source: Standard.

Proof. **Step S2.2: Bounded.** F is continuous on $W \supset K$, so $\|F\| \leq M$ on the compact K ((E5)).

Step S2.3: Contradiction hypothesis. If no L works, for every $k \in \mathbb{N}$ there are $x_k \neq y_k$ in K with $\|F(x_k) - F(y_k)\| > k\|x_k - y_k\|$; then $\|x_k - y_k\| < 2M/k \to 0$.

Step S2.4: Compactness. A subsequence $x_{k_j} \to p \in K$, hence also $y_{k_j} \to p$. By (E3) there are L_p and a neighbourhood N_p of p on which F is L_p-Lipschitz. For large j both points lie in N_p and $k_j > L_p$, which contradicts Step S2.3. □

Definition 2.5 (Uniform data)**.** Let F be C^1 on the open set W. A set $\Phi \subset E$ has *uniform data* (δ, M, L), with $\delta > 0$, $M, L \geq 0$, if for every $x \in \Phi$: $\overline{B}(x, \delta) \subset W$, $\|F\| \leq M$ on $\overline{B}(x, \delta)$ and F is L-Lipschitz on $\overline{B}(x, \delta)$. (Balls are convex, so by (E4) a bound $\|DF\| \leq L$ on $\overline{B}(x, \delta) \subset W$ gives the Lipschitz clause.)

Lemma 2.6 (Compact sets have uniform data)**.** *Let F be C^1 on the open set W and $K \subset W$ compact. Then K has uniform data (δ, M, L) for some $\delta > 0$, and $K_\delta = \{y : \operatorname{dist}(y, K) \leq \delta\} \subset W$.*

Proof. **Step S2.5: Choice of δ.** If $W = E$ put $\delta = 1$. Otherwise $x \mapsto \operatorname{dist}(x, E \setminus W)$ is continuous and positive on K, so has a minimum $2\delta > 0$ ((E5)); then $K_\delta \subset W$. K_δ is closed and bounded, hence compact.

Step S2.6: Data. Apply Lemma 2.4 to K_δ: $\|F\| \leq M$ and F is L-Lipschitz on $K_\delta \supset \overline{B}(x, \delta)$ for $x \in K$. □

Lemma 2.7 (Uniform data for angle-periodic fields)**.** *Let $E = \mathbb{R}^n \times \mathbb{R}^n$ (max norm), $U \subset \mathbb{R}^n$ open, $W = \mathbb{R}^n \times U$, F C^1 on W with $F(\theta + k, I) = F(\theta, I)$ for $(\theta, I) \in W$, $k \in \mathbb{Z}^n$. If $K \subset U$ is compact, then $\Phi = \mathbb{R}^n \times K$ has uniform data (δ, M, L) for some $\delta \in (0, 1]$.*

Proof. **Step S2.7: A compact cell.** Choose $\delta \in (0,1]$ with $K_\delta = \{I : \operatorname{dist}(I,K) \le \delta\} \subset U$ (as in Step S2.5). $C = [-1,2]^n \times K_\delta$ is compact and contained in W; let M, L_C be the constants of Lemma 2.4 for C.

Step S2.8: Translating a ball into the cell. Let $x = (\theta, I) \in \Phi$, $k = (\lfloor \theta_i \rfloor)_i \in \mathbb{Z}^n$ and $x' = x - (k,0) \in [0,1)^n \times K$. Since the norm is the max norm and $\delta \le 1$, $\overline{B}(x',\delta) \subset C$, and $\overline{B}(x,\delta) = \overline{B}(x',\delta) + (k,0) \subset W$. By periodicity $F(y + (k,0)) = F(y)$ on $\overline{B}(x',\delta)$, so $\|F\| \le M$ and F is L_C-Lipschitz on $\overline{B}(x,\delta)$. □

2.3. **Local existence.**

Lemma 2.8 (A priori short-time confinement)**.** *Let J be an interval, $t_0 \in J$, $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$ and $\alpha'(t) = F(\alpha(t))$ for all $t \in J$ (derivative two-sided; F is only assumed to be a function), and suppose $\|F\| \le M$ on $\overline{B}(x,\delta)$ with $M > 0$. Then $\|\alpha(t) - x\| \le M|t - t_0|$ for all $t \in J$ with $|t - t_0| < \delta/M$; in particular $\alpha(t) \in \overline{B}(x,\delta)$.*
Source: Standard.

Proof. **Step S2.9: Forward bootstrap.** Let $t \in J$, $t_0 < t < t_0 + \delta/M$ (the case $t < t_0$ follows by Lemma 2.3(b) and the case $t = t_0$ is trivial). Let $A = \{s \in [t_0, t] : \|\alpha(u) - x\| \le M(u - t_0)\ \forall u \in [t_0, s]\}$ and $s^* = \sup A$. $t_0 \in A$ and A is closed by continuity of α on J, so $s^* \in A$.

Step S2.10: Extension. If $s^* < t$, then $\|\alpha(s^*) - x\| \le M(s^* - t_0) < \delta$, so by continuity there is $\eta > 0$, $s^* + \eta \le t$, with $\alpha([s^*, s^* + \eta]) \subset \overline{B}(x,\delta)$. On this interval $\|\alpha'\| = \|F(\alpha)\| \le M$, so by (E4) $\|\alpha(u) - \alpha(s^*)\| \le M(u - s^*)$ and hence $\|\alpha(u) - x\| \le M(u - t_0)$ there, contradicting the definition of s^*. So $s^* = t$. □

Proposition 2.9 (Local existence with uniform time)**.** *Let F be C^1 on the open set W and Φ have uniform data (δ, M, L); put $M' = \max(M, 1)$ and $\varepsilon = \delta/M'$. Then for every $x \in \Phi$ and every $t_0 \in \mathbb{R}$ there is $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$, $\|\alpha(t) - x\| < \delta$ and* `IsSolIn` $F\, W\, (t_0 - \varepsilon, t_0 + \varepsilon)\, \alpha$.
Source: Standard (Picard–Lindelöf).

Proof. **Step S2.11: Picard–Lindelöf.** Apply (E1) with $a = \delta$, bound M', Lipschitz constant L and $T = \varepsilon$ ($M'T = \delta \le a$): there is α with $\alpha(t_0) = x$ and derivative $F(\alpha(t))$ within $[t_0 - \varepsilon, t_0 + \varepsilon]$ at each point of it. At points t of the open interval this is a two-sided derivative, since the closed interval is a neighbourhood of t.

Step S2.12: Staying in W. By Lemma 2.8 on $J = (t_0 - \varepsilon, t_0 + \varepsilon)$, $\|\alpha(t) - x\| \le M'|t - t_0| < \delta$, so $\alpha(t) \in \overline{B}(x,\delta) \subset W$. □

Theorem 2.10 (Local existence, both time directions)**.** *Let F be C^1 on the open set W, $x_0 \in W$, $t_0 \in \mathbb{R}$. There are $\varepsilon > 0$ and $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x_0$ and* `IsSolIn` $F\, W\, (t_0 - \varepsilon, t_0 + \varepsilon)\, \alpha$. *Moreover ε may be taken uniform on any compact $K \subset W$ (for all $x_0 \in K$).*
Source: Standard.

Proof. **Step S2.13: Reduction.** $\{x_0\}$ (resp. K) is compact; Lemma 2.6 gives uniform data and Proposition 2.9 the claim. □

2.4. **Uniqueness on every interval.**

Theorem 2.11 (Uniqueness on arbitrary intervals)**.** *Let F be C^1 on the open set W, $J \subset \mathbb{R}$ an interval (of any kind: open, closed, half-open, unbounded, or a point), $t_0 \in J$, and $y_1, y_2 : \mathbb{R} \to E$ with* `IsSolIn` $F\, W\, J\, y_i$ *and $y_1(t_0) = y_2(t_0)$. Then $y_1 = y_2$ on J. No confinement, boundedness or smallness of either solution is assumed. In particular, if y_1, y_2 are solutions on intervals J_1, J_2 and agree at one point of $J_1 \cap J_2$, they agree on $J_1 \cap J_2$ (an interval), and two solutions which agree at one time of J agree at all times of J.*

Source: Standard; used for BM11 Theorem 2.1 (uniqueness) and in the maximal-solution argument of BM11 p. 13.

Proof. **Step S2.14: A compact piece.** Let $t_1 \in J$, $t_1 > t_0$ ($t_1 < t_0$ is the mirror image, using the left-derivative form of (E2); $t_1 = t_0$ is the hypothesis). Then $[t_0, t_1] \subset J$, each y_i is continuous on it, so $K = y_1([t_0, t_1]) \cup y_2([t_0, t_1])$ is compact ((E5)) and $K \subset W$.

Step S2.15: Lipschitz on K. Lemma 2.4 gives L with F L-Lipschitz on K.

Step S2.16: Apply (E2). Both y_i are continuous on $[t_0, t_1]$, have right derivative $F(y_i(t))$ on $[t_0, t_1)$ (a two-sided derivative gives a right derivative) and take values in K; so $y_1(t_1) = y_2(t_1)$. □

2.5. **Maximal solutions and continuation.**

Lemma 2.12 (Extension to an open interval)**.** *Let F be C^1 on the open set W, $J \neq \emptyset$ an interval and* `IsSolIn` $F\, W\, J\, y$. *Then there are an open interval $J' \supset J$ and y' with $y' = y$ on J and* `IsSolIn` $F\, W\, J'\, y'$.

Proof. **Step S2.17: Right end.** If J has a largest element b, apply Theorem 2.10 at $x_0 = y(b) \in W$, $t_0 = b$: α on $(b - \varepsilon, b + \varepsilon)$. Put $y_1(s) = y(s)$ for $s \leq b$ and $y_1(s) = \alpha(s)$ for $s > b$. Then y_1 is a solution on $J_1 = J \cup (b, b + \varepsilon)$: at $t \in J$, $t < b$, and at $t \in (b, b + \varepsilon)$ it agrees with y, resp. α, near t; at $t = b$ it has derivative $F(y(b))$ within $(-\infty, b]$ (it equals y there) and within $[b, \infty)$ (it equals α there, as $\alpha(b) = y(b)$), so by (E6) it has the two-sided derivative. If J has no largest element, $J_1 = J$, $y_1 = y$.

Step S2.18: Left end. Apply Step S2.17 to the reversed solution $y_1(-\cdot)$ (Lemma 2.3(b)). The result has no endpoint, so is an open interval. □

Theorem 2.13 (Maximal solution)**.** *Let F be C^1 on the open set W and $x_0 \in W$. There are $T_- \in [-\infty, 0)$, $T_+ \in (0, +\infty]$ and $z : \mathbb{R} \to E$ with $z(0) = x_0$ and* `IsSolIn` $F\, W\, (T_-, T_+)\, z$, *such that every solution y with $y(0) = x_0$ on an interval $J \ni 0$ satisfies $J \subset (T_-, T_+)$ and $y = z$ on J. Then (T_-, T_+) is the* maximal interval *and $z|_{(T_-, T_+)}$ the* maximal solution*; both are unique. For initial time t_0 instead of 0 use Lemma 2.3(a).*
Source: BM11 p. 13 ("maximal time of existence"); standard.

Proof. **Step S2.19: The union.** Let $\mathcal{J}$ be the set of open intervals $J \ni 0$ carrying a solution y_J with $y_J(0) = x_0$; $\mathcal{J} \neq \emptyset$ by Theorem 2.10. The union I of $\mathcal{J}$ is an open interval $(T_-, T_+) \ni 0$ (a union of intervals containing 0).

Step S2.20: Gluing. For $t \in I$ choose $J \in \mathcal{J}$ with $t \in J$ and set $z(t) = y_J(t)$ (and $z(t) = x_0$ outside I). This is independent of J: for $J, J' \in \mathcal{J}$, $y_J = y_{J'}$ on $J \cap J' \ni 0$ by Theorem 2.11. Each $t \in I$ has an open neighbourhood $J \in \mathcal{J}$ on which $z = y_J$, so z is a solution on I (local nature, (iii) above).

Step S2.21: Maximality. If y is a solution on an interval $J \ni 0$ with $y(0) = x_0$, Lemma 2.12 gives an open $J' \supset J$ and a solution y' extending y, so $J' \in \mathcal{J}$, $J \subset J' \subset I$, and $y = y' = z$ on J (Theorem 2.11). Uniqueness of (T_-, T_+) and z is immediate from the maximality property applied twice. □

Theorem 2.14 (Continuation criterion, uniform form)**.** *Let F be C^1 on the open set W, $x_0 \in W$, (T_-, T_+) and z the maximal interval and solution of Theorem 2.13. Let $\Phi \subset E$ have uniform data (δ, M, L) (Definition 2.5). If $T_+ < \infty$ there is no $t_* \in [0, T_+)$ with $z(t) \in \Phi$ for all $t \in [t_*, T_+)$. Symmetrically, if $T_- > -\infty$ there is no $t_* \in (T_-, 0]$ with $z(t) \in \Phi$ for all $t \in (T_-, t_*]$. Equivalently: if the solution stays in Φ near the right (left) end of its maximal interval, that end is $+\infty$ ($-\infty$).*
Source: Standard; BM11 p. 13 ("this solution is defined for all time"), Pöschel 1993 p. 198 ("$T = T_*$").

Proof. **Step S2.22: Set-up.** Suppose $T_+ < \infty$ and $z(t) \in \Phi$ on $[t_*, T_+)$. Let $\varepsilon = \delta/\max(M, 1)$ from Proposition 2.9. Choose $t_1 \in [t_*, T_+)$ with $t_1 > T_+ - \varepsilon$.

Step S2.23: A solution through $z(t_1)$. Proposition 2.9 with $x = z(t_1) \in \Phi$ and $t_0 = t_1$ gives α on $(t_1 - \varepsilon, t_1 + \varepsilon)$ with $\alpha(t_1) = z(t_1)$. By Theorem 2.11 on $(t_1 - \varepsilon, t_1 + \varepsilon) \cap (T_-, T_+)$, $\alpha = z$ there.

Step S2.24: Gluing past T_+. Put $y = z$ on (T_-, T_+) and $y = \alpha$ on $[T_+, t_1 + \varepsilon)$; then $y = \alpha$ on $(t_1 - \varepsilon, t_1 + \varepsilon)$ and $y = z$ on (T_-, T_+), two open sets covering $(T_-, t_1 + \varepsilon)$ on each of which y is a solution. So y is a solution on $(T_-, t_1 + \varepsilon) \ni 0$ with $y(0) = x_0$, and $t_1 + \varepsilon > T_+$ contradicts Theorem 2.13. The left end follows by Lemma 2.3(b). □

Corollary 2.15 (Compact continuation)**.** *In the setting of Theorem 2.14, if $K \subset W$ is compact and $z(t) \in K$ for all $t \in [t_*, T_+)$ (some $t_* \in [0, T_+)$), then $T_+ = +\infty$; likewise on the left.*

Proof. Lemma 2.6 gives uniform data for K; apply Theorem 2.14. □

Corollary 2.16 (Closed sets, periodic fields)**.** *Let $E = \mathbb{R}^n \times \mathbb{R}^n$, $U \subset \mathbb{R}^n$ open, $W = \mathbb{R}^n \times U$, F C^1 on W with $F(\theta + k, I) = F(\theta, I)$ for $k \in \mathbb{Z}^n$. If the maximal solution through $x_0 \in W$ satisfies $z(t)_2 \in K$ for $t \in [t_*, T_+)$ with $K \subset U$ compact, then $T_+ = +\infty$ (likewise on the left). The angle components $z(t)_1$ may be unbounded.*

Proof. Lemma 2.7 gives uniform data for $\Phi = \mathbb{R}^n \times K$; apply Theorem 2.14. □

Remark 2.17 (What the closed-set variant really needs). Staying in a closed non-compact set $\Phi \subset W$ does *not* suffice in general (the solution of $\dot{x} = x^2$ on $E = \mathbb{R}$, $W = \mathbb{R}$, $x_0 = 1$ stays in the closed set $[1, \infty)$ and blows up at $T_+ = 1$). What suffices is the uniform data (δ, M, L) of Definition 2.5: a δ-neighbourhood of Φ inside W on which F is bounded and Lipschitz, uniformly. This holds for compact sets, for angle-periodic fields over a compact action set (Lemma 2.7), and for the Newtonian regions bounded away from collisions with bounded velocities (Corollary 2.27).

Theorem 2.18 (Confinement glue)**.** *Let F be C^1 on the open set W, $x_0 \in W$, $\tau > 0$, and $\Phi \subset E$ a set with uniform data (e.g. compact $\Phi \subset W$, or as in Lemma 2.7). Assume the* a priori confinement*: every solution y on an open interval $J \ni 0$ with $y(0) = x_0$ satisfies $y(t) \in \Phi$ for all $t \in J$ with $|t| \le \tau$. Then the maximal interval contains $[-\tau, \tau]$ in its interior, so z is a solution on $[-\tau, \tau]$ with $z(t) \in \Phi$ for $|t| \le \tau$, and every solution on $[-L, L]$ $(0 \le L \le \tau)$ with initial value x_0 equals z there.*
Source: BM11 p. 13 (the two-case argument "Therefore this solution is defined for all time"); Pöschel 1993 p. 198.

Proof. **Step S2.25: No exit before τ.** Suppose $T_+ \le \tau$. The maximal solution z on (T_-, T_+) satisfies, for $t \in [0, T_+)$ $(|t| < \tau)$, $z(t) \in \Phi$ by confinement. Theorem 2.14 gives $T_+ = \infty$, a contradiction. So $T_+ > \tau$, and likewise $T_- > -\tau$.

Step S2.26: Conclusion. $[-\tau, \tau] \subset (T_-, T_+)$ so z is a solution on $[-\tau, \tau]$ and $z(t) \in \Phi$ for $|t| \le \tau$ by confinement applied to z itself. Uniqueness is Theorem 2.11. □

2.6. First exit from an open set; barriers.

Lemma 2.19 (First exit)**.** *Let* `IsSolIn F W J z` *with J an interval, $0 \in J$, $V \subset E$ open and $z(0) \in V$. Put $A = \{t \in J : t \ge 0,\ z(t) \notin V\}$. Then exactly one of the following holds.*

- *(i) $A = \emptyset$: $z(t) \in V$ for all $t \in J$, $t \ge 0$.*
- *(ii) $A \ne \emptyset$. Then $t^* = \min A$ exists, $t^* > 0$, $t^* \in J$, $z(t) \in V$ for $t \in [0, t^*)$ and $z(t^*) \in \partial V = \overline{V} \setminus V$. The number t^* is the* first exit time *and equals $\sup\{t \in J : t \ge 0,\ z([0,t]) \subset V\}$; in particular t^* is finite and lies in J (t^* is "inside the solution's interval" and the solution is defined, continuous, at t^*).*

The same holds backwards in time (replace $z(t)$ by $z(-t)$, Lemma 2.3(b)). Only continuity of z on J is used; F and W play no role.
Source: Standard; used for BM11 p. 13 ("smallest time t_* such that $\omega(t_*) \in R_K$") and Pöschel 1993 p. 198–199 (stopping-time arguments).

Proof. **Step S2.27: Closedness.** z is continuous on J (consequence (ii)), $E \setminus V$ is closed, so A is closed in $J \cap [0, \infty)$.

Step S2.28: The minimum. If $A \neq \emptyset$, $t^* = \inf A \in [0, \infty)$; $t^* \in J$ since J is an interval containing 0 and a point of A that is $\geq t^*$. By closedness $t^* \in A$, i.e. $z(t^*) \notin V$. As $z(0) \in V$, $t^* > 0$.

Step S2.29: Boundary point. For $t \in [0, t^*)$, $t \notin A$, so $z(t) \in V$. By continuity $z(t^*) = \lim_{t \uparrow t^*} z(t) \in \overline{V}$, and $z(t^*) \notin V$; as V is open, $\overline{V} \setminus V = \partial V$.

Step S2.30: The supremum formula. $z([0, t]) \subset V$ iff $t < t^*$ (for $t \in J$, $t \geq 0$), so the supremum is t^*. □

Definition 2.20 (Invariant sets)**.** $S \subset E$ is *invariant* for (F, W) if for every interval J' and every y with `IsSolIn` $F\, W\, J'\, y$: if $y(t_1) \in S$ for one $t_1 \in J'$ then $y(t) \in S$ for all $t \in J'$. (Equilibria $\{x : F(x) = 0\}$ are invariant by Theorem 2.11; so is any union of such sets and any set $\{G = c\}$ for a first integral G.)

Corollary 2.21 (Barrier)**.** *Let F be C^1 on the open set W, V open, `IsSolIn` $F\, W\, J\, z$ with J an interval, $0 \in J$, $z(0) \in V$. Let $G \subset E$ with $z(J) \subset G$, and suppose $\partial V \cap G \subset S$ where S is invariant and $z(0) \notin S$ (e.g. G an energy level, Theorem 2.23, and $\partial V \cap G$ a union of invariant sets not meeting the initial point). Then $z(t) \in V$ for all $t \in J$ (both time directions).*

Proof. **Step S2.31: Forward.** If not, Lemma 2.19(ii) gives $t^* \in J$ with $z(t^*) \in \partial V \cap G \subset S$; invariance gives $z(0) \in S$, a contradiction.

Step S2.32: Backward. Apply Step S2.31 to $t \mapsto z(-t)$, a solution of $-F$ on $-J$; G, ∂V are unchanged and S is invariant for $-F$ as well (reversal maps solutions of F to solutions of $-F$ and back). □

2.7. **Hamiltonian layer.** Phase space $P = \mathbb{R}^n \times \mathbb{R}^n$, points $z = (\theta, I)$, max norm. X_H is the frozen `KAM.hamiltonianVectorField`, Definition 1.3 (convention of [K:`bp:p0:hamiltonian`]): $X_H(z) = \big((\partial_{I_i} H(z))_i, (-\partial_{\theta_i} H(z))_i\big)$, with $\partial_{I_i} H(z) = DH(z)(0, e_i)$ and $\partial_{\theta_i} H(z) = DH(z)(e_i, 0)$.

Lemma 2.22 (Regularity of X_H)**.** *Let $W \subset P$ be open and $H : P \to \mathbb{R}$ be C^{k+1} on W, $k \geq 0$. Then X_H is C^k on W; for $k \geq 1$ it is C^1 (hence locally Lipschitz) on W, and H is differentiable at every point of W.*
Source: Standard.

Proof. **Step S2.33: Derivative is C^k.** $DH : W \to \mathcal{L}(P, \mathbb{R})$ is C^k on the open W ((E6), `ContDiffOn.fderiv_of_isOpen`).

Step S2.34: Components. Each coordinate of X_H is $\pm$ the composition of DH with the continuous linear evaluation $\ell \mapsto \ell(0, e_i)$ or $\ell \mapsto \ell(e_i, 0)$; hence C^k, and a map into $\mathbb{R}^n \times \mathbb{R}^n$ with C^k coordinates is C^k. □

Theorem 2.23 (Energy conservation)**.** *Let $H : P \to \mathbb{R}$, $J \subset \mathbb{R}$ an interval and $z : \mathbb{R} \to P$ such that for every $t \in J$, H is differentiable at $z(t)$ and z has derivative $X_H(z(t))$ at t. Then $t \mapsto H(z(t))$ is constant on J. (No regularity of H beyond the stated pointwise differentiability, and no confinement.)*
Source: BM11 p. 13 and Pöschel 1993 p. 198 ("energy conservation", "$\Delta H = H(I(t), \theta(t))|_0^T$ vanishes"); standard.

Proof. **Step S2.35: Chain rule.** For $t \in J$, $\frac{d}{dt} H(z(t)) = DH(z(t))\big[X_H(z(t))\big]$ ((E6)).

Step S2.36: Cancellation. Write $z = z(t)$, $\ell = DH(z)$ and $X_H(z) = (a, b)$ with $a_i = \ell(0, e_i)$, $b_i = -\ell(e_i, 0)$. By linearity, $\ell(a, b) = \sum_i a_i\, \ell(e_i, 0) + \sum_i b_i\, \ell(0, e_i) = \sum_i \ell(0, e_i)\ell(e_i, 0) - \sum_i \ell(e_i, 0)\ell(0, e_i) = 0$.

Step S2.37: Constancy. Let $s < t$ in J. $H \circ z$ is continuous on $[s, t] \subset J$ and has derivative 0 at every point, so $H(z(t)) = H(z(s))$ ((E6)). □

Lemma 2.24 (Periodicity in the angles passes to solutions)**.** *Let $W \subset P$ be open with $W + (k, 0) = W$ for all $k \in \mathbb{Z}^n$, and $H : P \to \mathbb{R}$ with $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W$, $k \in \mathbb{Z}^n$. Then $X_H(\theta + k, I) = X_H(\theta, I)$ on W. Consequently, if* `IsSolIn` $X_H\, W\, J\, z$ *then* `IsSolIn` $X_H\, W\, J\, (z + (k, 0))$ *for every $k \in \mathbb{Z}^n$, and $H \circ (z + (k, 0)) = H \circ z$.*
Source: Standard.

Proof. **Step S2.38: Field.** Let $\tau_k(x) = x + (k, 0)$. On the open set W, $H \circ \tau_k = H$, so $DH = D(H \circ \tau_k) = DH \circ \tau_k$ at each point of W (the derivative of a translate; both sides are 0 if H is not differentiable, by the definition of `fderiv`). Evaluating on $(0, e_i)$ and $(e_i, 0)$ gives $X_H \circ \tau_k = X_H$ on W.

Step S2.39: Solutions. Lemma 2.3(c) with $v = (k, 0)$ applies to $F = X_H$. □

2.8. **The frozen solution predicates are instances.**

Proposition 2.25 (Track N and track I Hamiltonian predicates)**.** *Let $H : P \to \mathbb{R}$, $U \subset \mathbb{R}^n$ open, $W \subset P$ open with $\mathbb{R}^n \times U \subset W$, and H be C^2 on W. Put $W_U = \mathbb{R}^n \times U$. For every $J \subset \mathbb{R}$ and $z : \mathbb{R} \to P$,*

$$\texttt{IsSolutionOn}\, H\, U\, J\, z \iff \texttt{IsSolIn}\, X_H\, W_U\, J\, z \iff \texttt{IsHamiltonianSolutionOn}\, H\, U\, J\, z,$$

and X_H is C^1 on W_U. (The three predicates are literally the same formula up to the letter naming the action set, `KAM.Nekhoroshev.IsSolutionOn` *and* `KAM.Iso.IsHamiltonianSolutionOn`*.)*
Frozen Lean declaration: `KAM.Nekhoroshev.IsSolutionOn, KAM.Iso.IsHamiltonianSolutionOn`
Source: Ours (bookkeeping).

Proof. **Step S2.40: Membership.** $z(t)_2 \in U \iff z(t) \in W_U$ by definition of W_U.

Step S2.41: Differentiability is free. For $z(t) \in W_U \subset W$, H is C^2 on the open W, hence differentiable at $z(t)$ (Lemma 2.22); so the clause `DifferentiableAt` $\mathbb{R}\, H\, (z\, t)$ is implied by $z(t) \in W_U$, and the two predicates coincide. Smoothness of X_H: Lemma 2.22, $k = 1$, restricted to $W_U \subset W$. □

Proposition 2.26 (Newtonian predicates)**.** *Let $N \geq 2$, $G \in \mathbb{R}$,* `mass` $: Fin\, N \to \mathbb{R}$. *Let $E_N = (Fin\, N \to \mathbb{R}^2) \times (Fin\, N \to \mathbb{R}^2)$ (max norm),* $\ell =$ `planeLength`*,*

$$W_N = \{(q, v) : \ell(q_j - q_i) > 0\ \forall i \neq j\}, \qquad F_N(q, v) = \Big(v,\ \big(\textstyle\sum_{j \neq i} \frac{G\, \mathrm{mass}_j}{\ell(q_j - q_i)^3}(q_j - q_i)\big)_i\Big).$$

Then W_N is open, F_N is C^∞ on W_N, and for $Q, V : \mathbb{R} \to (Fin\, N \to \mathbb{R}^2)$ with $x(t) = (Q(t), V(t))$: `IsNewtonSolution` $G\, \mathrm{mass}\, Q\, V \iff$ `IsSolIn` $F_N\, W_N\, \mathbb{R}\, x$. *Likewise for* `IsFixedSolution` $G\, m_0\, m\, \varepsilon$ *with $W_N' = W_N \cap \{\ell(q_i) > 0\ \forall i\}$ and the field $F'(q, v) = (v, (-\frac{Gm_0}{\ell(q_i)^3} q_i + \varepsilon \sum_{j \neq i} \frac{Gm_j}{\ell(q_j - q_i)^3}(q_j - q_i))_i)$, and for* `IsRestrictedSolutionOn` $\mu\, J$ *with $W_\mu = \{(q, p) :$* `CollisionFree` $\mu\, (q, p)\}$ *and the total extension F_μ of* `restrictedVectorField` μ *by the same formula (with the values off W_μ irrelevant):* `IsRestrictedSolutionOn` $\mu\, J\, z \iff$ `IsSolIn` $F_\mu\, W_\mu\, J\, z$. *In all three cases F is C^∞ on the open set W and the general layer applies with $J = \mathbb{R}$ (resp. the given J).*
Frozen Lean declaration: `KAM.Planetary.IsNewtonSolution, KAM.Planetary.IsFixedSolution, KAM.Iso.IsRestrictedSolutionOn`

Source: Ours (bookkeeping); Fejoz Section 6.1, Newton equations as in the frozen text.

Proof. **Step S2.42: Open domain and smoothness.** $\ell(w)^2 = \texttt{planeSq}(w) = w_0^2 + w_1^2$ is a polynomial, so $\{\ell > 0\} = \{\texttt{planeSq} > 0\}$ is open, and $\ell = \sqrt{\cdot}$ is C^∞ where $\texttt{planeSq} > 0$. Hence $\ell(q_j - q_i)^{-3}$ is C^∞ on W_N and F_N is a finite sum of products of C^∞ maps. Same for F' and F_μ ($\texttt{radius} = \sqrt{\texttt{radiusSq}}$ with the shifts by $\texttt{primary}_{1,2}\mu$).

Step S2.43: Derivatives of pairs. A map into a finite product has derivative (a, b) at t iff each component has the corresponding derivative (`hasDerivAt_pi`, `HasDerivAt.prodMk` and converse). The first block of the frozen definition (distance positivity) is membership in W_N; the second block says Q has derivative $V(t)$; the third says V has derivative the force. Together they say exactly that x has derivative $F_N(x(t))$ at t. □

Corollary 2.27 (Newtonian continuation)**.** *In the setting of Proposition 2.26, let x be the maximal solution through $x_0 \in W_N$ (Theorem 2.13), $\rho > 0$, $\bar V \geq 0$ and $\Phi_{\rho,\bar V} = \{(q, v) : \ell(q_j - q_i) \geq \rho \ (i \neq j),\ |v| \leq \bar V\}$. If $T_+ < \infty$ then for every $\rho, \bar V$ and every $t_* \in [0, T_+)$ there is $t \in [t_*, T_+)$ with $x(t) \notin \Phi_{\rho,\bar V}$ (a finite-time singularity needs near-collisions or unbounded velocities; positions may be unbounded). The same for F' with the additional condition $\ell(q_i) \geq \rho$, and for F_μ with $\ell \geq \rho$ from both primaries and $|q|, |p| \leq \bar V$.*

Proof. **Step S2.44: Uniform data.** Let $\delta = \rho/8$. If $|y - x| \leq \delta$ (max norm) with $x \in \Phi_{\rho,\bar V}$ then each $\ell(q'_j - q'_i) \geq \rho - 2\sqrt{2}\,\delta \geq \rho/2$, so $y \in W_N$ and $\overline{B}(x, \delta) \subset W_N$. On this ball $\ell \geq \rho/2$ and $|v'| \leq \bar V + \delta$; the entries of F_N and DF_N are polynomials in v', the differences $q'_j - q'_i$ (Euclidean length $\leq \ell(q'_j - q'_i)$) and the inverse powers ℓ^{-3}, ℓ^{-5} ($D(w/\ell^3) = \ell^{-3}I - 3\ell^{-5}ww^\top$), hence bounded by a constant $C(N, G, \text{mass}, \rho, \bar V)$ *independent of the position of x*. By (E4) on the convex ball, F_N is C-Lipschitz there. So $\Phi_{\rho,\bar V}$ has uniform data.

Step S2.45: Apply. Theorem 2.14 gives the claim; the other two fields are identical in structure (for F_μ the extra terms are linear in (q, p) and bounded on $|q|, |p| \leq \bar V$). □

2.9. **The track-N statements.** Throughout, $R, s > 0$, $h : \mathbb{C}^n \to \mathbb{C}$ with $\texttt{IsAnalyticIntegrable}\, R\, s\, h$, $\varepsilon \geq 0$ and $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ with $\texttt{IsAnalyticPerturbation}\, R\, s\, \varepsilon\, f$ (Definitions 1.11, 1.12), $H = H_{h,f}$ (Definition 1.4), $W_s = \mathbb{R}^n \times B(0, R + s) \subset P$ (open; $B(0, R + s) = \mathcal{B}_s(R)$ by Remark 1.9), and $U \subset B(0, R + s)$ open (in N-HEAD-C, $U = B(0, R)$).

Lemma 2.28 (Input from Part 1: H is smooth on W_s)**.** *$H_{h,f}$ is C^∞ (in particular C^2) on the open set W_s, $X_H = X_{h+f}$ on $W_s \cap (\mathbb{R}^n \times \mathbb{R}^n)$, and $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W_s$, $k \in \mathbb{Z}^n$.*
Source: Part 1 forward label [N:`bpN:p1:holo-real`]*; the two ingredients are* [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:hamiltonian-real`]*; periodicity is Definition 1.12(ii).*

Proof. **Step S2.46: Regularity.** The function $G(\theta, I) = h(I) + f(\theta, I)$ is holomorphic on the open set $D_s(R) \supset W_s$ (real points (θ, I) with $I \in \mathcal{B}_s(R) = B(0, R)_s \cap \mathbb{R}^n$ lie in $T_s \times B(0, R)_s$), hence C^∞ in the real sense ([K:`bp:p1:holomorphic-regularity`]); it is real on W_s by Definition 1.11(iv) and 1.12(iii). So $H = \operatorname{Re} G = G$ on W_s is the restriction of C^∞ map to a real subspace: C^∞ on W_s; $X_H = X_G$ at real points is [K:`bp:p1:hamiltonian-real`]. (Packaged as [N:`bpN:p1:holo-real`].)

Step S2.47: Periodicity. $f(\theta + k, I) = f(\theta, I)$ for real θ, $I \in \mathcal{B}_s(R)$, and h depends on I only. □

Theorem 2.29 (X_H is C^1, hence locally Lipschitz)**.** *$X_{H_{h,f}}$ is C^1 on $W_s \supset \mathbb{R}^n \times B(0, R)$; consequently it is Lipschitz on every compact subset of W_s, and on every set $\mathbb{R}^n \times K$ with $K \subset B(0, R+s)$ compact there are δ, M, L as in Lemma 2.7.*
Source: BM11 p. 13 (implicit in "the solution"); standard.

Proof. **Step S2.48: Smoothness.** Lemma 2.28 and Lemma 2.22 ($k = 1$).

Step S2.49: Lipschitz. Lemma 2.4 on compact sets, Lemma 2.7 (with $U = B(0, R+s)$, using the periodicity of X_H from Lemma 2.24 and Lemma 2.28) for the cylinders. □

Theorem 2.30 (Local existence and uniqueness for `IsSolutionOn`)**.** *For every $t_0 \in \mathbb{R}$ and $z_0 \in \mathbb{R}^n \times U$ there are $\epsilon > 0$ and $y : \mathbb{R} \to P$ with $y(t_0) = z_0$ and $\mathtt{IsSolutionOn}\, H\, U\, (t_0 - \epsilon, t_0 + \epsilon)\, y$ (both time directions). If J is any interval, $t_0 \in J$ and y_1, y_2 satisfy $\mathtt{IsSolutionOn}\, H\, U\, J\, y_i$ with $y_1(t_0) = y_2(t_0)$, then $y_1 = y_2$ on J.*
Source: BM11 p. 13 and Theorem 2.1 p. 5; Picard–Lindelöf (external fact (E1)).

Proof. **Step S2.50: Reduction.** Proposition 2.25 (with $W = W_s$, H C^2 there by Lemma 2.28) turns $\mathtt{IsSolutionOn}\, H\, U$ into $\mathtt{IsSolIn}\, X_H\, (\mathbb{R}^n \times U)$ with X_H C^1 on $\mathbb{R}^n \times U$.

Step S2.51: Existence, uniqueness. Theorem 2.10 and Theorem 2.11. □

Theorem 2.31 (Energy conservation for `IsSolutionOn`)**.** *If $\mathtt{IsSolutionOn}\, H\, U\, J\, z$ for an interval J (here H may be any function $P \to \mathbb{R}$ and U any set), then $H(z(t)) = H(z(t'))$ for $t, t' \in J$. In particular, for $H = H_{h,f}$ and $J = [0, T]$ the quantity $\Delta H = H(z(t))|_0^T$ of Pöschel's Lemma 1 vanishes.*
Source: Pöschel 1993 p. 198 ("by energy conservation ... ΔH vanishes"; page read); BM11 p. 13.

Proof. **Step S2.52: Direct.** The definition gives differentiability of H at every $z(t)$ and $z'(t) = X_H(z(t))$; Theorem 2.23. □

Theorem 2.32 (Maximal solutions and continuation)**.** *Let $z_0 = (\theta_0, I_0)$ with $I_0 \in U$. There is a unique maximal interval $(T_-, T_+) \ni 0$ and a solution z with $z(0) = z_0$ and $\mathtt{IsSolutionOn}\, H\, U\, (T_-, T_+)\, z$, such that every y with $y(0) = z_0$ and $\mathtt{IsSolutionOn}\, H\, U\, J\, y$ on an interval $J \ni 0$ has $J \subset (T_-, T_+)$ and $y = z$ on J (Theorem 2.13 through Proposition 2.25). If $K \subset U$ is compact and $z(t)_2 \in K$ for all $t \in [t_*, T_+)$ for some $t_* \in [0, T_+)$, then $T_+ = +\infty$; likewise on the left. The angle components $z(t)_1$ may be unbounded: no obstruction arises because X_H is 1-periodic in θ.*
Source: BM11 p. 13 ("From this estimate we can deduce that the solution $I(t)$ belongs to some compact ball ... this solution is defined for all time"); Pöschel 1993 p. 198 ("$T = T_*$").

Proof. **Step S2.53: General statements.** By Proposition 2.25, $\mathtt{IsSolutionOn}\, H\, U\, J\, y \iff \mathtt{IsSolIn}\, X_H\, (\mathbb{R}^n \times U)\, J\, y$ and X_H is C^1 on $W = \mathbb{R}^n \times U$; maximality is Theorem 2.13.

Step S2.54: Periodic cylinder. X_H is 1-periodic in θ on W (Lemmas 2.24, 2.28). Corollary 2.16 yields $T_+ = \infty$. □

Theorem 2.33 (Uniqueness on closed intervals)**.** *For $L \geq 0$ and y_1, y_2 with $y_1(0) = y_2(0)$ and $\mathtt{IsSolutionOn}\, H\, U\, [-L, L]\, y_i$ one has $y_1 = y_2$ on $[-L, L]$. In particular a solution on $[-L, L]$ agrees with every solution on a larger interval containing $[-L, L]$ and the same initial value (clause (d) of Theorem 1.19, where $U = B(0, R)$, $L \in [0, \tau]$, y arbitrary and z the exhibited solution). No confinement of y is assumed: Gronwall is replaced by Theorem 2.11, whose Lipschitz constant comes from the compact set $y_1([-L, L]) \cup y_2([-L, L])$.*
Source: Standard (Gronwall); clause (d) of the frozen N-HEAD-C.

Proof. **Step S2.55: Apply.** $J = [-L, L]$, $t_0 = 0$ in Theorem 2.30 (uniqueness part). □

Theorem 2.34 (Confinement glue)**.** *Let $\theta_0 \in \mathbb{R}^n$, $I_0 \in U$, $\tau > 0$ and $\rho > 0$ with $\overline{B}(I_0, \rho) \subset U$ (e.g. $|I_0| < R/2$, $\rho \leq R/4$, $U = B(0, R)$). Assume that for every open interval $J \ni 0$ and every y with $y(0) = (\theta_0, I_0)$ and $\mathtt{IsSolutionOn}\, H\, U\, J\, y$ one has $|y(t)_2 - I_0| \leq \rho$ for all $t \in J$ with $|t| \leq \tau$. Then there is z with $z(0) = (\theta_0, I_0)$, $\mathtt{IsSolutionOn}\, H\, U\, [-\tau, \tau]\, z$ and $|z(t)_2 - I_0| \leq \rho$ for $|t| \leq \tau$;*

and for every $L \in [0,\tau]$ and every y with $y(0) = z(0)$ and `IsSolutionOnHU` $[-L,L]\,y$, *$y = z$ on $[-L,L]$. This is the assembly of clauses (a)–(d) of Theorem 1.19 from an a priori estimate (Part 10); $\rho = C\varepsilon^{1/(2n)}$ there.*
Source: BM11 p. 13 (two cases, "Therefore this solution is defined for all time"); Pöschel 1993 p. 198.

Proof. **Step S2.56: Compact confinement set.** $K = \overline{B}(I_0,\rho)$ is compact and $\subset U$; $\Phi = \mathbb{R}^n \times K$ has uniform data by Lemma 2.7 (with $W = \mathbb{R}^n \times U$, $F = X_H$ periodic by Lemma 2.24).

Step S2.57: General glue. By Proposition 2.25 the hypothesis reads: every general solution on an open $J \ni 0$ with this initial value stays in Φ for $|t| \le \tau$. Theorem 2.18 gives z on $[-\tau,\tau]$ in Φ, i.e. $|z(t)_2 - I_0| \le \rho$, and uniqueness on $[-L,L]$ (it is also Theorem 2.33). □

2.10. Label index for tracks I and P.

`bpN:p3:gen-sol`, `gen-symmetry`	predicate; time shift, reversal, space translation
`gen-lip`, `gen-uniform`, `gen-uniform-compact`, `gen-uniform-periodic`	Lipschitz and uniform data
`gen-short`, `gen-uniform-exist`, `gen-local`	local existence, uniform time
`gen-uniqueness`	uniqueness on all intervals, no confinement
`gen-extend`, `gen-maximal`	maximal solutions
`gen-continuation`, `-compact`, `-periodic`, `gen-confinement`	continuation, closed sets with uniform bounds, confinement glue
`gen-exit`, `gen-invariant`, `gen-barrier`	first exit, barrier
`ham-smooth`, `ham-energy`, `ham-periodic`	Hamiltonian layer
`ham-reduction`, `newton-reduction`, `newton-continuation`	frozen predicates of I, N, P
`holo-input`, `lipschitz`, `local-exist`, `energy`, `continuation`, `uniqueness-closed`, `confinement-glue`	track N specializations